

Trey Harrison

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Education

University of Arizona - W.A. Franke Honors College

Expected Dec 2027

B.S. Computer Science, Minors in Mathematics & Artificial Intelligence

- Accelerated Master's Program, M.S. Computer Science - expected May 2029
- Relevant Coursework: Data Structures & Algorithms, Vector Calculus, Discrete Math, Systems Programming in C, Linear Algebra

Experience

Software Developer - NASA Aspera Mission

May 2026 - Present

- Developed spacecraft geometry and ephemeris tooling using NASA's SPICE toolkit, generating kernels to compute satellite position, attitude, and ground-tracking geometry from orbital data
- Built attitude and clock kernels for the mission's pointing instrument, converting camera-matrix data into SPICE-native frames to target-visibility analysis
- Contributed to a team of 100+ doctors, researchers, and developers to study and analyze data from the diffuse extreme-ultraviolet (EUV) emission from hot gas in the circumgalactic medium

Student Developer - Center for Digital Humanities

May 2025 - Present

- Traveled with a team of developers to Curacao to perform LIDAR scanning of cemeteries for digital recreation, as well as on-site volumetric recordings to create holographic assets
- Collaborated with cultural institutions including the Smithsonian and delivered technical instruction on volumetric-capture and AR to graduate-level University of Arizona courses
- Build and maintained WebAR viewers using Three.js and AWS S3/Cloudfront asset pipelines; reverse engineered a proprietary volumetric format and migrated 200+ AR projects onto university infrastructure after shutdown of Niantic 8th Wall

Undergraduate Teaching Assistant - Department of Computer Science

January 2026 - May 2026

- Graded programming projects and tests, wrote test cases, and evaluated student success for a 100+ Data Structures and Algorithms course

Projects

TEMI Robot Messaging Service - Android Studio | Java | AWS SES

- Designed a robust end-to-end voice message delivery system using AWS SES supporting offline fallback and local storage

Quantum Computing AI Chatbot - React | Node.js | Hugging Face | PostgreSQL

- Created a full stack RAG architecture indexing Q# documentation and injecting relevant contextual information via semantic search into a Mistral-7B model using llama-cpp

Skills, Awards, Certifications, and Scholarships

Programming: Java, JavaScript, Python

Frameworks & Tools: React, Node.js, AWS SES, Firebase, Blender, 8th Wall, Git, PostgreSQL

AI/XR: Gaussian Splatting, Photogrammetry, RAG Pipelines, Volumetric Capture, Llama-cpp, ChromaDB

Awards, Certifications, and Scholarships

Honors Research and Creative Inquiry Award, W.A. Franke Honors College

September 2025

Innovation Award, University of Arizona

April 2025

Dean's List with Distinction, University of Arizona

December 2024